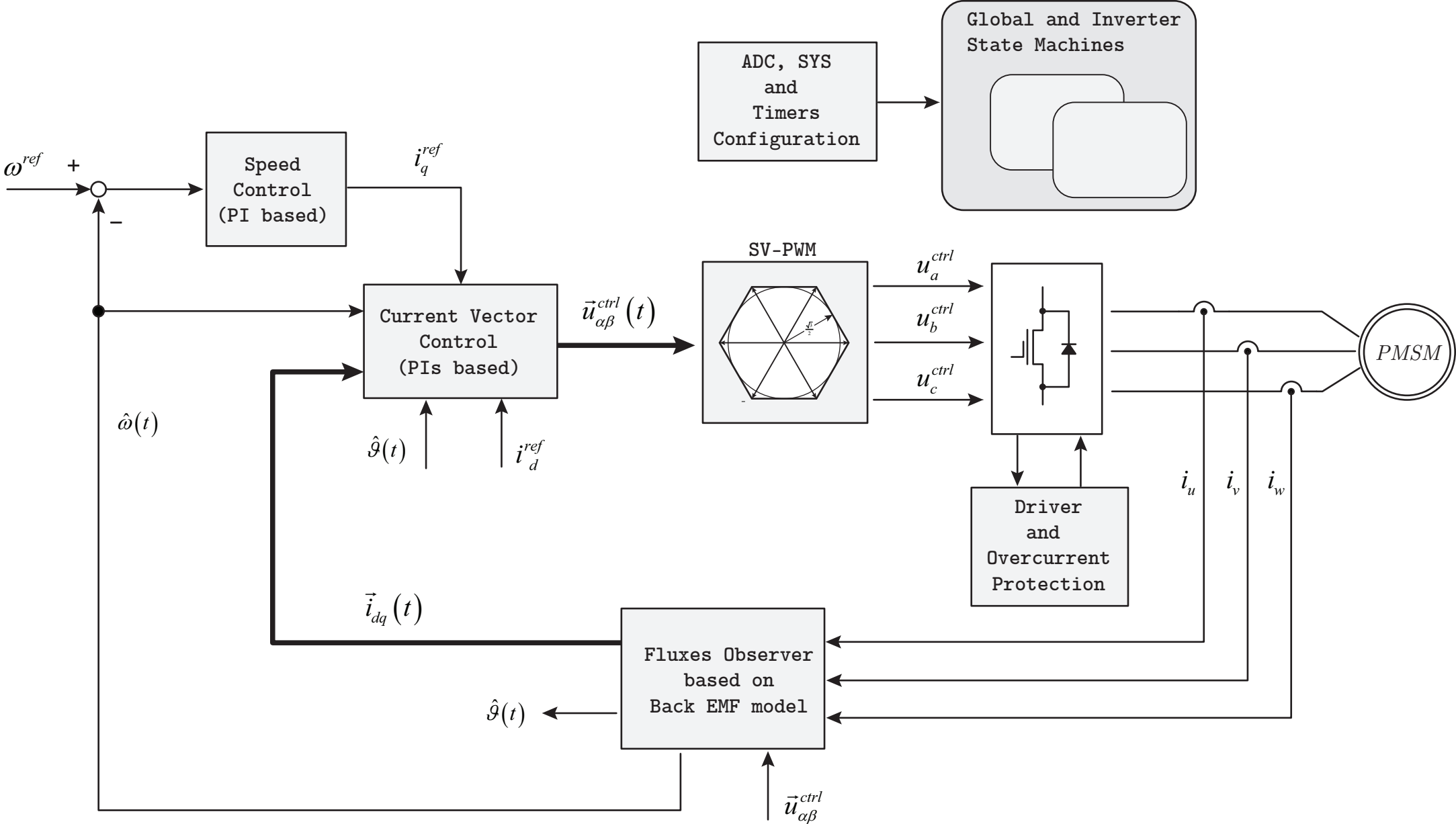
ADC, SYS
and
Timers
Configuration
Global and Inverter
State Machines
ω^{ref}
+
–
Speed
Control
(PI based)
i_q^{ref}
SV-PWM
u_a^{ctrl}
u_b^{ctrl}
u_c^{ctrl}
Current Vector
Control
(PIs based)
$\vec{u}_{\alpha\beta}^{ctrl}(t)$
PMSM
$\hat{\omega}(t)$
$\hat{\vartheta}(t)$
i_d^{ref}
i_u
i_v
i_w
Driver
and
Overcurrent
Protection
$\vec{i}_{dq}(t)$
Fluxes Observer
based on
Back EMF model
$\hat{\vartheta}(t)$
$\vec{u}_{\alpha\beta}^{ctrl}$